University of Utah Spring 2026

MATH 3210-004 Selected Solutions to PSet Problems

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Contents

1 Introduction

In this document you may find solutions (proofs or examples) to selected problems from problem sets. As a general rule of thumb, it is more likely that this document will be updated with the solutions to problems that were graded on accuracy.

Throughout this class your solutions need not match the level of polish, precision and detail of the below solutions. What's most important is grasping the underlying concepts and steadily improving your problem-solving skills as well as your proof-writing and presentation skills.

Most solutions will have follow-up exercises. Just as in the problem sets, these follow-up exercises are stated in a neutral language, and determining whether or not the statement is correct is part of the exercise. A follow-up exercise here may later show up in an exam.

2 PSet 1, Problem 3

Let X and Y be sets. Let R be a relation from X to Y. Then

(i) The relation $\mathcal{R} = \{ (A, B) \mid B = R(A) \}$ from $\mathcal{P}(X)$ to $\mathcal{P}(Y)$ is the graph of a function.

(ii) For any $A, B \subseteq X$, $R(A \cup B) = R(A) \cup R(B)$.

(iii) For any $A, B \subseteq X$, $R(A \cap B) = R(A) \cap R(B)$.

(iv) For any $A \subseteq X$, $R\left(A^{\complement}\right) = R(A)^{\complement}$.

(v) For any $A, B \subseteq X$, $R(A \setminus B) = R(A) \setminus R(B)$.

We'll study each statement separately. The first statement is true:

Claim: The relation $\mathcal{R} = \{ (A, B) \mid B = R(A) \}$ from $\mathcal{P}(X)$ to $\mathcal{P}(Y)$ is the graph of a function. ⌟

Proof: We need to verify that the relation $\mathcal{R}$ from $\mathcal{P}(X)$ to $\mathcal{P}(Y)$ defined by

$$\mathcal{R} = \{(A, B) | B = R(A)\} \subseteq \mathcal{P}(X) \times \mathcal{P}(Y)$$

is the graph of a function with domain $\mathcal{P}(X)$. For $A \in \mathcal{P}(X)$, by definition $R(A) \in \mathcal{P}(Y)$, so that indeed $\mathrm{dom}(\mathcal{R}) = \mathcal{P}(X)$. Next let $(A_1, B_1), (A_2, B_2) \in \mathcal{R}$. Suppose $A_1 = A_2$. We need to show that $B_1 = B_2$. As $(A_1, B_1), (A_2, B_2) \in \mathcal{R}$, by the definition of $\mathcal{R}$, we have that

$$B_1 = R(A_1), \quad B_2 = R(A_2).$$

To show the equality of two sets, it suffices to show that either one is a subset of the other one. So let $y \in B_1 = R(A_1)$. Then for some $x \in A_1$, $(x, y) \in R$. But as $A_1 = A_2$, $x \in A_2$ also, and since $(x, y) \in R$, we have that $y \in R(A_2) = B_2$. Thus $B_1 \subseteq B_2$. Interchanging the roles of B_1 and B_2 in this argument, we immediately obtain also $B_1 \supseteq B_2$, whence $B_1 = B_2$. In other words, $\mathcal{R}$ satisfies the abstract vertical line test, and consequently is the graph of a function from $\mathcal{P}(X)$ to $\mathcal{P}(Y)$.

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The second statement is also true:

Claim: For any $A, B \subseteq X$, $R(A \cup B) = R(A) \cup R(B)$.

⌟

Proof: Let $A, B \subseteq X$. Then as $A \subseteq A \cup B$, $R(A) \subseteq R(A \cup B)$. Indeed, if $y \in R(A)$, there is an $x \in A$ such that $(x, y) \in R$. But $A \subseteq A \cup B$, so that $x \in A \cup B$, thus $y \in R(A \cup B)$. Similarly $R(B) \subseteq R(A \cup B)$. Then $R(A) \cup R(B) \subseteq R(A \cup B)$.

For the other direction, let $y \in R(A \cup B)$. Then there is an $x \in A \cup B$ such that $(x, y) \in R$. By definition $x \in A$ or $x \in B$, so that $y \in R(A)$ or $y \in R(B)$, that is, $R(A \cup B) \subseteq R(A) \cup R(B)$.

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The remaining three statements are false in general. To prove that a given statement is false, it suffices to provide one counterexample to the statement. For instance, the third statement is: "The image of the intersection of two subsets is equal to the intersection of the images". The logical negation of this is the statement "The image of the intersection of two subsets may be different than the intersection of the images". Using logical symbols, we may succintly write:

$$\neg[\,\forall A, B \in \mathcal{P}(X) : R(A \cap B) = R(A) \cap R(B)\,]$$
$$\equiv \exists A, B \in \mathcal{P}(X) : R(A \cap B) \neq R(A) \cap R(B).$$

We'll prove that the given statement is false, that is, the negation

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of the given statement if true. As this latter is an existential statement, one example for it is sufficient; the example for the negation is a counterexample for the given statement.

Claim: The image of the intersection of two subsets under a relation is not necessarily equal to the intersection of the images.

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Proof: Consider $X = [0,1] = Y$, $A = [0,1/4]$, $B = [3/4,1]$, $f : X \to Y$, $x \mapsto 1/2$, that is, f is the constant function that is constantly $1/2$. Then $A \cap B = \emptyset$, hence $f(A \cap B) = \emptyset$, while $f(A) = f(B) = \{1/2\}$, so that $f(A) \cap f(B) = \{1/2\}$.

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To obtain a correct version of the third statement, either we can weaken the conclusion, or strengthen the hypotheses[1].

Claim: The following version of the third statement with weakened conclusion is true:

$$\forall A, B \in \mathcal{P}(X) : R\,(A \cap B) \subseteq R(A) \cap R(B).$$

⌟

Proof: Let $A, B \subseteq X$. As $A \cap B \subseteq A$, $R(A \cap B) \subseteq R(A)$. Similarly as $A \cap B \subseteq B$, $R(A \cap B) \subseteq R(B)$. Thus $R(A \cap B) \subseteq R(A) \cap R(B)$.

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Claim: The following version of the third statement with an additional hypothesis is true:

[1]These are instances of logical perturbations of the original statement given; you do not need to submit such corrected versions, though trying to come up with such corrected versions will help you understand the concepts and prepare you for the exams better.

$$\begin{aligned} & [\,\forall (x_1, y_1), (x_2, y_2) \in R : y_1 = y_2 \implies x_1 = x_2\,] \\ \iff \quad & [\,\forall A, B \in \mathcal{P}(X) : R\,(A \cap B) = R(A) \cap R(B)\,]. \end{aligned}$$

⌟

Proof: ($\Longrightarrow$) Suppose R satisfies

$$\forall (x_1, y_1), (x_2, y_2) \in R : y_1 = y_2 \implies x_1 = x_2;$$

one might call this the **injectivity condition** (aka **abstract horizontal line test**) for relations. Let $A, B \subseteq X$, and $y \in R(A) \cap R(B)$. Then there are $a \in A$ and $b \in B$ such that $(a, y), (b, y) \in R$. By the injectivity condition, this forces that $a = b$, and consequently that this element is in $A \cap B$. Thus $y \in R(A \cap B)$, so that $R(A) \cap R(B) \subseteq R(A \cap B)$. Since $R(A) \cap R(B) \supseteq R(A \cap B)$ for any relation R by the previous claim, we have $R(A) \cap R(B) = R(A \cap B)$.

($\Longleftarrow$) Suppose R fails the injectivity condition, so that there are $x_1, x_2 \in X$ and $y \in Y$ such that $x_1 \neq x_2$ and $(x_1, y), (x_2, y) \in R$. Define $A = \{x_1\}, B = \{x_2\}$. Then $A \cap B = \emptyset$, so $R(A \cap B)$, but $y \in R(A) \cap R(B)$, that is, $R(A \cap B) \subsetneq R(A) \cap R(B)$.

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Claim: The image of the absolute complement of a subset under a relation need not be equal to the absolute complement of the image.

⌟

Proof: We may repurpose our previous counterexample to establish this claim also. Put $X = Y = [0, 1]$, $A = [0, 1/4]$, $f : X \to Y, x \mapsto 1/2$. Then $A^\complement =]1/4, 1]$, $f(A^\complement) = \{1/2\}$, but $f(A)^\complement = \{1/2\}^\complement$, hence in fact we have that the two subsets are disjoint.

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Claim: The image of the relative complement of two subsets under a relation need not be equal to the relative complement of the images.

⌟

Proof: Recalling that $A \setminus B = A \cap B^{\complement}$, the counterexample we gave earlier serves as a counterexample for this claim also. Indeed, put $X = Y = [0, 1]$, $A = [0, 1/4]$, $B = [3/4, 1]$, $f : X \to Y$, $x \mapsto 1/2$. Then $A \cap B^{\complement} = [0, 1/4]$, hence $f(A \setminus B) = \{1/2\}$; on the other hand $f(A) = f(B) = \{1/2\}$, so that $f(A) \setminus f(B) = \emptyset$.

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2.1 Follow-up Exercises

(i) The second and the (correct versions) of the last third statements hold for arbitrary collections $\mathcal{A} \subseteq \mathcal{P}(X)$.

(ii) The fourth and fifth statement has correct versions (one involving a "subset of" relation, and one involving the injectivity condition; similar to the correct versions of the third statement).

(iii) If $f : X \to Y$ is a function, then the preimage relation f^{-1} satisfies the injectivity condition[2]. Consequently, taking preimages commutes with unions, intersections and relative complements.

[2]Recall that f^{-1} really is an abbreviation for $\text{graph}(f)^{-1}$, and f^{-1} is not necessarily a function from Y to X.

3 PSet 1, Problem 6

Let X and Y be two sets. A function from X to Y is **surjunctive** if it is either surjective or non-injective. Then

(i) X is finite iff any function from X to X is surjunctive.

(ii) Y has exactly one element iff any function from X to Y is surjunctive.

The first claim is correct without any modifications, while the second statement is not quite correct as given. Still, a minor adjustment suffices to obtain a correct version. We start with the first claim.

For this claim, part of the challenge really is to make intuitive sense of what it means for a set to be finite[3]. From a set theoretical point of view, this calls for a rigorous definition, however we'll keep the notion of finiteness of sets at the naive level[4]. Thus for us, a set X is **finite** if it is either the emptyset or if it can be represented as

$$X = \{x_1, x_2, ..., x_n\}$$

for some positive integer n. Similarly, a set X is **infinite** if for any positive integer n, there is an element $x_n \in X$ with the property that $x_i \neq x_j$ for $i \neq j$. With these definitions, for instance, it is true that saying that a set is not infinite is equivalent to saying that the set is finite, although proving this equivalence rigorously is surprisingly cumbersome! As we are really focused toward calculus, we'll not investigate this issue any further.

Claim: Let X be a set. Then X is finite iff for any $f \in F(X;X)$, f is

[3] In *this* sentence, we use the word "intuitive" intuitively and not in a technical sense. Throughout this course we'll almost never use "intuition" in a technical sense.

[4] Indeed, it is valid in the grand scheme of things to take the claim we set out to prove as the *definition* of what it means for a set to be finite.

surjunctive.

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Proof: Recall that "iff" stands for a necessary and sufficient condition. Thus we'll need to prove two implications.

($\Rightarrow$) First suppose $X = \{x_1, x_2, ..., x_n\}$ is a finite set with exactly n elements and let $f : X \to X$ be an arbitrary function. We claim that f is surjunctive. That is, either f is surjective, or it is non-injective. To prove this, we may assume f fails to have one of these properties and show that it must have the other property. To that end, suppose f is injective (so that it fails to be non-injective). We need to show that f is surjective.

Suppose for a contradiction that there is a $j \in \{1, 2, ..., n\}$ such that $x_j \notin \mathrm{im}(f)$. But then im(f) has at most $n - 1$ distinct elements. On the other hand, by the injectivity of f, for any $i, j \in \{1, 2, ..., n\}$ with $i \neq j$, $f(x_i) \neq f(x_j)$, so that im(f) has at least n distinct elements, a contradiction[5].

($\Leftarrow$) For the converse, we'll argue contrapositively. Suppose X is infinite, so that we have for any positive integer n, an element $x_n \in X$ with the property that $x_i \neq x_j$ for $i \neq j$. We may then define a self-function of X as follows:

$$f : X \to X, x \mapsto \begin{cases} x_{n+1}, & \text{if } \exists n \in \mathbb{Z}_{\geq 1} : x = x_n \\ x, & \text{else} \end{cases}.$$

Thus f is the function that shifts the index of the specified indexed elements by one, and acts as identity on $X \setminus \{x_n | n \in \mathbb{Z}_{\geq 1}\}$. Then by construction, $x_1 \notin \mathrm{im}(f)$, however f is injective, so that f fails to be surjunctive.

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[5]Alternatively, one can refer to the pigeonhole principle to derive a contradiction.

The second claim is not correct in general. Indeed, say $X = \{x_1, x_2, x_3\}$, $Y = \{y_1, y_2\}$. Then there are no injections from X to Y, thus any function $X \to Y$ is automatically surjunctive. The best we can do is to verify one side of the equivalence:

Claim: Let X and Y be two sets. If Y has exactly one element then any function $f : X \to Y$ is surjunctive.

⌟

Proof: If $X = \emptyset$, then there are no functions $X \to Y$, so we may assume that $X \neq \emptyset$. Put $Y = \{y\}$ and let $f : X \to Y$ be function. Then f must be surjective, as there is at least one $x \in X$, and $f(x)$ is an element of Y, and consequently $f(x) = y$. Then by definition, f being surjective implies it being surjunctive.

⌟

3.1 Follow-up Exercises

(i) Let Y be a set. Then Y has exactly one element iff for any set X, any function in $F(X; Y)$ is surjunctive.

4 PSet 2, Problem 1

$$\forall n \in \mathbb{Z}_{\geq 1} : \sum_{k=1}^{n} k^3 = \frac{1}{4}n^4 + \frac{1}{2}n^3 + \frac{1}{4}n^2.$$

The statement given is correct without any modifications. We'll give both an induction proof and a direct proof. For brevity let us first set up some notation. Put

$$S : \mathbb{Z}_{\geq 1} \times \mathbb{Z}_{\geq 0} \to \mathbb{R}, (n, p) \mapsto \sum_{k=1}^{n} k^p.$$

Thus we have, for any n, $S(n, 0) = n$ and $S(n, 1) = \dfrac{n(n+1)}{2}$ (we proved this latter formula in class).

Claim:

$$\forall n \in \mathbb{Z}_{\geq 1} : S(n, 3) = \frac{1}{4}n^4 + \frac{1}{2}n^3 + \frac{1}{4}n^2.$$

⌟

Proof (By Induction)**:** For the base step, we need to verify that the statement holds for $n = 1$. Indeed; $S(1, 3) = 1$ and

$$\left[\frac{1}{4}n^4 + \frac{1}{2}n^3 + \frac{1}{4}n^2\right]\Bigg|_{n=1} = \frac{1}{4} + \frac{1}{2} + \frac{1}{4} = 1.$$

For the induction step, let $n_* \in \mathbb{Z}_{\geq 1}$ be arbitrary and suppose the statement holds for $n = n_*$, that is,

$$S(n_*, 3) = \frac{1}{4}n_*^4 + \frac{1}{2}n_*^3 + \frac{1}{4}n_*^2. \qquad (\star)$$

We need to show that the statement holds for $n = n_* + 1$. We'll start with the polynomial function on the RHS and through some algebraic identities rewrite it in terms of $S(n_*, 3)$; specifically we'll make use of the binomial formula:

$$\begin{aligned}
&\frac{1}{4}(n_* + 1)^4 + \frac{1}{2}(n_* + 1)^3 + \frac{1}{4}(n_* + 1)^2 \\
&= \frac{1}{4}(n_*^4 + 4n_*^3 + 6n_*^2 + 4n_* + 1) \\
&\quad + \frac{1}{2}(n_*^3 + 3n_*^2 + 3n_* + 1) \\
&\quad + \frac{1}{4}(n_*^2 + 2n_* + 1) \\
&= \left(\frac{1}{4}n_*^4 + \frac{1}{2}n_*^3 + \frac{1}{4}n_*^2\right) \\
&\quad + \left(\frac{4}{4}\right) n_*^3 \\
&\quad + \left(\frac{6}{4} + \frac{3}{2}\right) n_*^2 \\
&\quad + \left(\frac{4}{4} + \frac{3}{2} + \frac{2}{4}\right) n_* \\
&\quad + \left(\frac{1}{4} + \frac{1}{2} + \frac{1}{4}\right) \\
&\overset{\star}{=} S(n_*, 3) + \left(n_*^3 + 3n_*^2 + 3n_* + 1\right) \\
&= S(n_*, 3) + (n_* + 1)^3 \\
&= S(n_* + 1, 3),
\end{aligned}$$

where in the equality with $\star$ we used the induction hypothesis. Thus by the principle of mathematical induction, we are done.

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Proof (Direct Proof)**:** For a direct proof, we'll relate the sum of p-th powers to the sums of lesser powers. Let $k, p \in \mathbb{Z}_{\geq 1}$[6]. Then

[6] When reading what follows, it might be helpful to take $p = 3$ to parse the manipulations.

$$(k+1)^{p+1} = \sum_{i=0}^{p+1} \binom{p+1}{i} k^i = k^{p+1} + (p+1)k^p + \sum_{i=0}^{p-1} \binom{p+1}{i} k^i$$

$$\implies k^p = \frac{(k+1)^{p+1} - k^{p+1}}{p+1} - \frac{1}{p+1} \sum_{i=0}^{p-1} \binom{p+1}{i} k^i,$$

where for $\alpha, \beta \in \mathbb{Z}_{\geq 1}$ with $\alpha \geq \beta$,

$$\binom{\alpha}{\beta} = \frac{\alpha!}{\beta!(\alpha-\beta)!} = \frac{\alpha(\alpha-1)\cdots(\alpha-\beta+1)}{\beta(\beta-1)\cdots 2} \in \mathbb{Z}.$$

Hence for $n \in \mathbb{Z}_{\geq 1}$ we have

$$\begin{aligned}
S(n,p) &= \sum_{k=1}^{n} k^p \\
&= \sum_{k=1}^{n} \left(\frac{(k+1)^{p+1} - k^{p+1}}{p+1} - \frac{1}{p+1} \sum_{i=0}^{p-1} \binom{p+1}{i} k^i \right) \\
&= \frac{1}{p+1} \sum_{k=1}^{n} [(k+1)^{p+1} - k^{p+1}] - \frac{1}{p+1} \sum_{i=0}^{p-1} \binom{p+1}{i} \left(\sum_{k=1}^{n} k^i \right) \\
&= \frac{1}{p+1} \sum_{k=1}^{n} [(k+1)^{p+1} - k^{p+1}] - \frac{1}{p+1} \sum_{i=0}^{p-1} \binom{p+1}{i} S(n,i) \\
&= \frac{(n+1)^{p+1} - 1}{p+1} - \frac{1}{p+1} \sum_{i=0}^{p-1} \binom{p+1}{i} S(n,i).
\end{aligned}$$

We have that $S(n,0) = n$ by definition. Thus plugging in $p = 1$ in this latter formula, we obtain

$$S(n,1) = \frac{(n+1)^2 - 1}{2} - \frac{1}{2}n = \frac{n^2+n}{2},$$

(thus giving another direct proof for the formula for the sum of first powers). Next we plug in p = 2 to obtain:

$$\begin{aligned}
S(n,2) &= \frac{(n+1)^3-1}{3} - \frac{1}{3}\sum_{i=0}^{1}\binom{3}{i}S(n,i) \\
&= \frac{(n+1)^3-1}{3} - \frac{1}{3}\left(S(n,0)+3S(n,1)\right) \\
&= \frac{(n+1)^3-1}{3} - \frac{1}{3}\left(S(n,0)+3S(n,1)\right) \\
&= \frac{(n+1)^3-1}{3} - \frac{1}{3}S(n,0) - S(n,1) \\
&= \frac{(n+1)^3-1}{3} - \frac{n}{3} - \frac{n^2+n}{2} \\
&= \frac{1}{3}n^3 + \frac{1}{2}n^2 + \frac{1}{6}n.
\end{aligned}$$

Finally taking p = 3 and substituting the formulas we obtained so far, we end up with the formula for the cubes of integers; we omit the routine algebraic manipulations.

⌟

Note that while the direct proof seems longer and more complicated than the proof by induction, it does not require an explicit formula and hence proves more than what is claimed, and further it signifies a method for a direct proof for any p.

4.1 Follow-up Exercises

(i) For any $p \in \mathbb{Z}_{\geq 1}$, $S(\bullet, p)$ is a polynomial of degree $p+1$ with rational coefficients.

(ii) Let $\alpha(p), \mu(p)$ be the minimal numbers of rational number additions and rational number multiplications one would need to execute to compute all the coefficients of $S(\bullet, p)$, respectively, for p an arbitrary positive integer.

(a) Compute $\alpha(p)$ and $\mu(p)$ for $p \in \{0, 1, 2\}$.

(b) Find two functions $\hat{\alpha}$ and $\hat{\mu}$ that can be written as the composition of finitely many elementary functions with the property that for some $C \in \mathbb{R}_{>0}$ and some $P \in \mathbb{Z}_{\geq 1}$:

$$\forall p \in \mathbb{Z}_{\geq P}: \quad \alpha(p) \leq C\hat{\alpha}(p), \quad \mu(p) \leq C\hat{\mu}(p).$$

(c) Find the best functions $\hat{\alpha}$ and $\hat{\beta}$ that have the properties in the previous part.

5 PSet 2, Problem 2

Let $A \subseteq \mathbb{R}$ be nonempty and bounded. Then

(i) The set of all upper bounds of A is $[\sup(A), \infty[$.

(ii) The set of all lower bounds of A is $]-\infty, \inf(A)]$.

(iii) The smallest closed interval containing A is

$$[\inf(A), \sup(A)].$$

(iv) The largest open interval contained in A is

$$]\inf(A), \sup(A)[.$$

Let $A \subseteq \mathbb{R}$. First let us set some notation for convenience. Define by $\mathcal{U}(A)$, $\mathcal{L}(A)$ the sets of all upper and lower bounds of A, respectively:

$$\mathcal{U}(A) = \{U \in \mathbb{R} | \forall a \in A : a \leq U\},$$
$$\mathcal{L}(A) = \{L \in \mathbb{R} | \forall a \in A : L \leq a\}.$$

We first establish that the first three statements are correct without modification:

Claim: Let $A \subseteq \mathbb{R}$ be nonempty and bounded, then

(i) $\mathcal{U}(A) = [\sup(A), \infty[\subseteq \mathbb{R}$,

(ii) $\mathcal{L}(A) =]-\infty, \inf(A)] \subseteq \mathbb{R}$,

(iii) The smallest closed interval containing A is $[\inf(A), \sup(A)] \subseteq \mathbb{R}$.

(iv) The largest open interval contained in A is $]\inf(A), \sup(A)[$ iff A is an interval.

⌟

Proof: We are given that A is bounded and nonempty, hence we have that neither $\mathcal{U}(A)$ nor $\mathcal{L}(A)$ are empty. In particular, by the completeness axiom we have that the minimum of $\mathcal{U}(A)$ as well as the maximum of $\mathcal{L}(A)$ exist; sup(A) and inf(A) are *defined* to be these real numbers.

(i) and (ii) Note that if U is an upper bound, then any number larger than U too is an upper bound, and similarly one can replace a lower bound by an even lower number. Thus we have:

$$\mathcal{U}(A) = \bigcup_{U \in \mathcal{U}(A)} [U, \infty[\ , \qquad \mathcal{L}(A) = \bigcup_{L \in \mathcal{L}(A)}]-\infty, L]. \qquad (\star)$$

In addition, if U is an upper bound of A, then $\sup(A) \leq U$, that is, $U \in [\sup(A), \infty[$, so that $\mathcal{U}(A) = [\sup(A), \infty[$. Similarly if L is a lower bound of A, L must be in $]-\infty, \inf(A)]$, so that $\mathcal{L}(A) \subseteq]-\infty, \inf(A)]$. This concludes the proof of the first two parts.

(iii) We may argue as follows for the third part: By definition inf(A) is a lower bound for any element in A and similarly sup(A) is an upper bound for any element in A, thus $A \subseteq [\inf(A), \sup(A)]$. If $[\lambda, \rho] \subseteq \mathbb{R}$ were a closed and bounded interval containing A, then λ would be a lower bound and ρ would be an upper bound for A; hence by definition $\lambda \leq \inf(A)$ and $\sup(A) \leq \rho$. That is $[\inf(A), \sup(A)] \subseteq [\lambda, \rho]$. In other words, $[\inf(A), \sup(A)]$ indeed is the smallest closed interval containing A.

⌟

Finally we study the last statement. In general the last statement

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fails, indeed, for instance for $A = \{0, 1\}$, a set with exactly two elements, $1/2$ is in $]0, 1[=]\inf(A), \sup(A)[$[7].

Still, if we assume that A is an interval to begin with, the fourth statement becomes true. Recall that the definition we are using for A to be an interval is that it satisfies the following property:

$$\forall x, y \in A, \forall t \in [0, 1] : (1 - t)x + ty \in A.$$

Claim: Let $A \subseteq \mathbb{R}$ be nonempty and bounded, then A is an interval iff the largest open interval contained in A is $]\inf(A), \sup(A)[$.

⌟

Proof: Let A be nonempty and bounded. We have two cases: either $\inf(A) = \sup(A)$ xor $\inf(A) \lneq \sup(A)$. In the first case, we have that A is a singleton, and consequently it does not contain any nonempty open interval, while it trivially satisfies the condition for being an interval, so that there is nothing to do. Suppose therefore that $\inf(A) \lneq \sup(A)$.

($\Rightarrow$) Suppose A is a nonempty and bounded interval and let $\mu \in]\inf(A), \sup(A)[$. Then μ fails to be a lower bound as well as an upper bound of A, by the definitions of infimum and supremum, respectively. Thus there are $x, y \in A$ such that

$$\inf(A) \leq x \lneq \mu \lneq y \leq \sup(A).$$

Since A is an interval and $\mu = (1 - t_*)x + t_*y$ for $t_* = \frac{\mu - x}{y - x} \in]0, 1[$, $\mu \in A$. Therefore $]\inf(A), \sup(A)[\subseteq A$.

Let $]\lambda, \rho[$ be an arbitrary open interval contained in A. WLOG we may assume that this interal is nonempty and take $\mu \in]\lambda, \rho[$. Then $\mu \in A$ and hence $\mu \in [\inf(A), \sup(A)]$. It must be the case that $\mu \gneq \inf(A)$, for otherwise, $\frac{\lambda + \inf(A)}{2}$, a number strictly less than $\inf(A)$, would

[7] For the purposes of problem set grading, concluding here would have been sufficient.

be in A, a contradiction. Similarly it must be the case that $\mu \lneq \sup(A)$, whence $]\lambda, \rho[\subseteq] \inf(A), \sup(A)[$, so that indeed $] \inf(A), \sup(A)[$ is the largest open interval contained in A.

($\Leftarrow$) We'll argue by contradiction. Suppose A is bounded and nonempty, that $] \inf(A), \sup(A)[$, but that it fails to be an interval. Then there are $x, y \in A$ and $t_* \in]0, 1[$ such that $x \lneq y$ and $(1 - t_*)x + t_*y \notin A$. But then

$$(1 - t_*)x + t_*y \in]x, y[\subseteq] \inf(A), \sup(A)[\subseteq A,$$

a contradiction.

⌟

We make a final remark to contextualize this problem: We had discussed the "clamp" heuristic for supremum and infimum in class, and the problem really is a formalization of this heuristic: increasing L in $]-\infty, L]$ corresponds to tightening the lower clamp, decreasing U in $[U, \infty[$ corresponds to tighthening the upper clamp, and the numbers L and U that correspond to the tighthest possible configuration are the infimum and supremum, respectively. The heuristic works for clamping from without. While there are physical devices that clamp from within, they are different in principle than the abstract notions on infimum and supremum, whence the fourth part is incorrect unless we have additional information regarding the nature of the set we are trying to clamp down on.

6 PSet 2, Problem 3

Let $A, B \subseteq \mathbb{R}$. Then

(i) $\sup(A \cup B) = \max\{\sup(A), \sup(B)\}$

(ii) $\inf(A \cup B) = \min\{\inf(A), \inf(B)\}$.

We separate cases as usual: either at least one of A and B is empty, xor neither A nor B is empty.

The $\boxed{A = \emptyset \text{ or } B = \emptyset}$ **Case** WLOG, we may assume $A \neq \emptyset = B$. Then $A \cup B = A$ and

$$\sup(A \cup B) = \sup(A) > -\infty = \sup(\emptyset) = \sup(B),$$
$$\inf(A \cup B) = \inf(A) < \infty = \inf(\emptyset) = \inf(B).$$

Since the maximum of an extended real number R and $-\infty$ is R by definition, the first part follows. Similarly since the minimum of an extended real number R and ∞ is R by definition the second part is also true.

The $\boxed{A \neq \emptyset \neq B}$ **Case** For the first part, WLOG we may assume that $\sup(A) \geq \sup(B)$. Thus it suffices to show that $\sup(A) = \sup(A \cup B)$. First note that as $A \subseteq A \cup B$, by the monotonicity of supremum we have

$$-\infty \lneq \sup(B) \leq \sup(A) \leq \sup(A \cup B) \leq \infty.$$

Either $A \cup B$ is unbounded from above, xor it is bounded from above. If $A \cup B$ is unbounded from above, then at least one of A or

B would have to be unbounded from above. If A is unbounded from above, then $\sup(A) = \infty = \sup(A \cup B)$ concludes the proof. If B is unbounded from above, then $\infty = \sup(B) \leq \sup(A) \leq \sup(A \cup B) \leq \infty$, hence again $\sup(A) = \sup(A \cup B)$ (and consequently A is unbounded from above also).

Now suppose $A \cup B$ is bounded from above; then the same is true for both A and B, and we have

$$-\infty \lneq \sup(B) \leq \sup(A) \leq \sup(A \cup B) \lneq \infty.$$

Say $\sup(A) \lneq \sup(A \cup B)$. Then $\sup(A)$ is not an upper bound for $A \cup B$, so that there is an $x \in A \cup B$ such that

$$\sup(A) < x.$$

Since x exceeds the least upper bound for A, $x \notin A$, thus it must be the case that $x \in B$. But then

$$x \leq \sup(B) \leq \sup(A) < x;$$

a contradiction.

One can argue analogously for the statement involving infima. Alternatively one can argue as follows[8]:

$$\begin{aligned} \inf(A \cup B) &= -\sup(-(A \cup B)) \\ &= -\sup((-A) \cup (-B)) \\ &= -\max\{\sup(-A)\,,\ \sup(-B)\} \\ &= -\max\{-\inf(A)\,,\ -\inf(B)\} \\ &= \min\{\inf(A)\,,\ \inf(B)\}, \end{aligned}$$

[8]You should check these steps if they are not clear!

where $-C = \{-c \in \mathbb{R} | c \in C\}$ for $C \subseteq \mathbb{R}$ a subset.

We have thus proved that indeed the statements in the problem are correct without any modifications:

Claim: $\forall A, B \subseteq \mathbb{R}$:

(i) $\sup(A \cup B) = \max\{\sup(A), \sup(B)\}$

(ii) $\inf(A \cup B) = \min\{\inf(A), \inf(B)\}$.

⌟

6.1 Follow-up Exercises

(i) $\forall \mathcal{A} \subseteq \mathcal{P}(\mathbb{R})$:

(a) $\sup\left(\bigcup_{A\in\mathcal{A}} A\right) = \sup_{A\in\mathcal{A}} \sup(A)$,

(b) $\inf\left(\bigcup_{A\in\mathcal{A}} A\right) = \inf_{A\in\mathcal{A}} \inf(A)$.

(ii) $\forall \mathcal{A} \subseteq \mathcal{P}(\mathbb{R})$:

(a) $\sup\left(\bigcap_{A\in\mathcal{A}} A\right) = \inf_{A\in\mathcal{A}} \sup(A)$,

(b) $\inf\left(\bigcap_{A\in\mathcal{A}} A\right) = \sup_{A\in\mathcal{A}} \inf(A)$.

7 PSet 3, Problem 2

Let $x_\bullet, y_\bullet$ be two convergent sequences of real numbers with limits $x_\infty, y_\infty \in \mathbb{R}$ respectively:

$$\lim_{n\to\infty} x_n = x_\infty, \qquad \lim_{n\to\infty} y_n = y_\infty.$$

Then

(i) $\lim_{n\to\infty}(x_n + y_n) = x_\infty + y_\infty$.

(ii) For any real number α, $\lim_{n\to\infty}(\alpha x_n) = \alpha x_\infty$.

(iii) $\lim_{n\to\infty}(x_n y_n) = x_\infty y_\infty$.

(iv) $\lim_{n\to\infty} \max\{x_n, y_n\} = \max\{x_\infty, y_\infty\}$.

All these limit laws are indeed correct without any modifications. The last statement involving maxima is slightly trickier than the rest, so we'll study it separately.

Claim: Let $x_\bullet \to x_\infty, y_\bullet \to y_\infty$ be two convergent sequences of real numbers. Then

(i) $\lim_{n\to\infty}(x_n + y_n) = x_\infty + y_\infty$.

(ii) For any real number α, $\lim_{n\to\infty}(\alpha x_n) = \alpha x_\infty$.

(iii) $\lim_{n\to\infty}(x_n y_n) = x_\infty y_\infty$.

⌟

For statements like these, we'll need to take an arbitrary $\varepsilon \in \mathbb{R}_{>0}$ and find some finite time $N \in \mathbb{Z}_{\geq 0}$ such that after this time, the distance between the position at time n (n being a time later than N) and the claimed target at most ε. Perhaps surprisingly, instead of starting with the ε, often it's more practical to start from the *end of the proof*. In other words, first with start with the distance we are trying to control, and then using some inequalities we establish that the distance we are trying to control can be controlled via the distances we are given control of. With all this said, a proof might go like this:

Proof: To be more concrete, let us start indexing the times of the sequences involved from 0[9].

(i) For the first part, we have, for any n,

$$|(x_n + y_n) - (x_\infty + y_\infty)| \leq |x_n - x_\infty| + |y_n - y_\infty| \qquad (\star)$$

by the triangular inequality. We know that each of the two summands on the RHS can be made arbitrarily small, and so this will suffice.

Let $\varepsilon \in \mathbb{R}_{>0}$. Then there are two times $N_1, N_2 \in \mathbb{Z}_{\geq 0}$ such that for any $n \in \mathbb{Z}_{\geq 0}$:

$$\begin{aligned} n \geq N_1 &\implies |x_n - x_\infty| < \varepsilon/2, \\ n \geq N_2 &\implies |y_n - y_\infty| < \varepsilon/2. \end{aligned}$$

But then, putting $N = \max\{N_1, N_2\}$, we have that for any $n \in \mathbb{Z}_{\geq 0}$, if $n \geq N$, then both of these bounds hold simultaneously. Thus for $n \geq N$, by $(\star)$, we have

$$|(x_n + y_n) - (x_\infty + y_\infty)| < \varepsilon.$$

(ii) We have for any $n \in \mathbb{Z}_{\geq 0}$,

[9] Ultimately at what time one starts indexing does not matter for the purposes of this proof, as the claim is all about the *future asymptotics*, insofar as the indices of different sequences are compatible.

$$|(\alpha x_n) - (\alpha x_\infty)| = |\alpha||x_n - x_\infty| \leq (|\alpha| + 1)|x_n - x_\infty|. \qquad (\dagger)$$

Let $\varepsilon \in \mathbb{R}_{>0}$. Then $\varepsilon/(|\alpha| + 1) \in \mathbb{R}_{>0}$ also, and hence there is a time $N \in \mathbb{Z}_{\geq 0}$ such that for any $n \in \mathbb{Z}_{\geq 0}$:

$$n \geq N \implies |x_n - x_\infty| < \varepsilon/(|\alpha| + 1).$$

Using this latter bound together with ($\dagger$), we obtain that for any $n \geq N$,

$$|(\alpha x_n) - (\alpha x_\infty)| < \varepsilon.$$

(iii) This one is slightly more tricky. Whenever one tries to bound distances between "terms with factors", a good method is to "inject" some choice cross terms[10]. Thus we have, for any n,

$$\begin{aligned}
&|(x_n y_n) - (x_\infty y_\infty)| \\
&= |x_n y_n - x_n y_\infty + x_n y_\infty - x_\infty y_\infty| \\
&\leq |x_n y_n - x_n y_\infty| + |x_n y_\infty - x_\infty y_\infty| \\
&= |x_n||y_n - y_\infty| + |x_n - x_\infty||y_\infty| \\
&= |x_n - x_\infty + x_\infty||y_n - y_\infty| + |x_n - x_\infty||y_\infty| \\
&\leq (|x_n - x_\infty| + |x_\infty|)|y_n - y_\infty| + |x_n - x_\infty||y_\infty|. \qquad (\$)
\end{aligned}$$

Let $\varepsilon \in \mathbb{R}_{>0}$. Then $\min\{1, \varepsilon/2(|y_\infty| + 1)\} \in \mathbb{R}_{>0}$ also[11], and since $x_\bullet \to x_\infty$, there is an $N_1 \in \mathbb{Z}_{\geq 0}$ such that for any $n \in \mathbb{Z}_{\geq 0}$:

[10] Here by "inject" I mean adding and subtracting the same term and then using the triangular inequality.

[11] As you are reading this proof, at this point you *should not* worry about why this number is relevant, and keep reading. For now, this is merely a possible error margin that depends on ε already chosen. See also the next footnote at the end of the proof.

$$n \geq N_1 \implies |x_n - x_\infty| < \min\{1, \varepsilon/2(|y_\infty| + 1)\}.$$

Further, $\varepsilon/2(|x_\infty| + 1) \in \mathbb{R}_{>0}$ too, and since $y_\bullet \to y_\infty$, there is an $N_2 \in \mathbb{Z}_{\geq 0}$ such that for any $n \in \mathbb{Z}_{\geq 0}$:

$$n \geq N_2 \implies |y_n - y_\infty| < \varepsilon/2(|x_\infty| + 1).$$

Combining these two inequalities with ($), we have that, putting $N = \max\{N_1, N_2\}$, for any $n \in \mathbb{Z}_{\geq 0}$,

$$n \geq N \implies |(x_n y_n) - (x_\infty y_\infty)| < \varepsilon.$$[12]

⌟

Claim ((iv))**:** Let $x_\bullet \to x_\infty, y_\bullet \to y_\infty$ be two convergent sequences of real numbers. Then

$$\lim_{n\to\infty} \max\{x_n, y_n\} = \max\{x_\infty, y_\infty\}.$$

⌟

We'll provide two separate proofs of this claim; one is more straight-

[12]How did I come up with the error margins for $x_\bullet$ and $y_\bullet$, as functions of ε, so that when the dust settled I obtained the nice ε bound for what I ultimately wanted to control? You'll see such bizarre expressions in the middle of proofs in analysis quite often; and the answer almost always is this: the author of the proof again worked *backwards*, and did some calculations not worthy of explanation (supposedly). In such cases it's best to not dwell too much on the specific choices of numbers and to keep reading the proof (as mentioned in the previous footnote). For instance in this case I was really looking at the RHS in ($). I want this term to be less than ε. As this is the sum of two terms, I decided to simply bound both summands by $\varepsilon/2$ each. The second summand has a factor that may or may not be 0, so I scaled $\varepsilon/2$ by dividing it by $|y_\infty| + 1 \in \mathbb{R}_{>0}$. But then I realized I'm going to need to bound the first term by $\varepsilon/2$ also, and this scaled version of $\varepsilon/2$ is too cumbersome to manipulate algebraically, hence I simply took the minimum of it with 1, in case the first supposed bound is relatively large etc.. There are many other bounds that would have worked, and ultimately none of this is really necessary. Instead you can simply bound the RHS by some function of ε that decays to zero as ε does. For instance in this case it's much more straightforward to bound the RHS by a quadratic polynomial in ε that has a zero at $\varepsilon = 0$ (such a polynomial decays like ε^2, which for ε small enough, is even smaller than ε.). There are contexts where one *needs* to be careful about sharp bounds (for instance if the problem was asking statements regarding the rates of convergences), but often there is a fair amount of flexibility with epsilon management.

forward but requires that one first establishes an inequality regarding maxima, while the other one is less straightforward but does not use any technology. For the first proof we first establish a lemma:

Lemma: For any $\alpha, \overline{\alpha}, \beta, \overline{\beta} \in \mathbb{R}$,

$$|\max\{\alpha, \overline{\alpha}\} - \max\{\beta, \overline{\beta}\}| \leq \max\{|\alpha - \beta|, |\overline{\alpha} - \overline{\beta}|\}.$$

⌟

Proof: Wlog, we may assume $\alpha \leq \overline{\alpha}$ and $\beta \leq \overline{\beta}$. Then

$$|\max\{\alpha, \overline{\alpha}\} - \max\{\beta, \overline{\beta}\}| = |\overline{\alpha} - \overline{\beta}| \leq \max\{|\alpha - \beta|, |\overline{\alpha} - \overline{\beta}|\}.$$

⌟

Proof (of Claim (iv), using Lemma)**:** We have that, by the lemma, for any $n \in \mathbb{Z}_{\geq 0}$:

$$|\max\{x_n, y_n\} - \max\{x_\infty, y_\infty\}| \leq \max\{|x_n - x_\infty|, |y_n - y_\infty|\}. \qquad (\spadesuit)$$

Let $\varepsilon \in \mathbb{R}_{>0}$. Then there are $N_1, N_2 \in \mathbb{Z}_{\geq 0}$ such that for any $n \in \mathbb{Z}_{\geq 0}$:

$$\begin{aligned} n \geq N_1 &\implies |x_n - x_\infty| < \varepsilon, \\ n \geq N_2 &\implies |y_n - y_\infty| < \varepsilon. \end{aligned}$$

Consequently, putting $N = \max\{N_1, N_2\}$, by ($\spadesuit$), we have that for any $n \in \mathbb{Z}_{\geq 0}$,

$$n \geq N \implies |\max\{x_n, y_n\} - \max\{x_\infty, y_\infty\}| < \varepsilon.$$

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Proof (of Claim (iv), without using Lemma)**:** Wlog, we may assume that $x_\infty \le y_\infty$. Then either $x_\infty \lneq y_\infty$ xor $x_\infty = y_\infty$.

The $\boxed{x_\infty \lneq y_\infty}$ **Case** In this case we have that $(y_\infty - x_\infty)/2 \in \mathbb{R}_{>0}$. Thus, using this positive number as the error margin, we have that there is a time $N \in \mathbb{Z}_{\ge 0}$ such that[13] for any $n \in \mathbb{Z}_{\ge 0}$,

$$\begin{aligned} & n \ge N \\ & \implies |x_n - x_\infty| < (y_\infty - x_\infty)/2 \\ & \qquad \text{and } |y_n - y_\infty| < (y_\infty - x_\infty)/2. \end{aligned}$$

This implies[14] that

$$n \ge N \implies x_n \lneq y_n.$$

Consequently, if $n \ge N$, we have

$$|\max\{x_n, y_n\} - \max\{x_\infty, y_\infty\}| = |y_n - y_\infty|. \qquad (\clubsuit)$$

Let $\varepsilon \in \mathbb{R}_{>0}$. Then there is an $N' \in \mathbb{Z}_{\ge 0}$ such that for any $n \in \mathbb{Z}_{\ge 0}$,

$$n \ge N' \implies |y_n - y_\infty| < \varepsilon.$$

Putting $N'' = \max\{N, N'\}$, using this last inequality together with $(\clubsuit)$, we have that for any $n \in \mathbb{Z}_{\ge 0}$,

[13] Here we are fast-forwarding the procedure of first finding two separate times for the two sequences to be sufficiently near their respective limits and then taking the maximum of these two times so as to guarantee simultaneity.

[14] Draw a picture, or rearrange the above inequalities!

$$n \geq N'' \implies |\max\{x_n, y_n\} - \max\{x_\infty, y_\infty\}| < \varepsilon.$$

The $\boxed{x_\infty = y_\infty}$ **Case** For clarity denote by L the common limit. Then both $x_\bullet$ and $y_\bullet$ converges to L. Let $\varepsilon \in \mathbb{R}_{>0}$. Then there is an $N \in \mathbb{Z}_{\geq 0}$ such that for any $n \in \mathbb{Z}_{\geq N}$,

$$|x_n - L| < \varepsilon,$$
$$|y_n - L| < \varepsilon.$$

Rearranging these inequalities, we obtain that for any $n \in \mathbb{Z}_{\geq N}$,

$$L - \varepsilon < x_n < L + \varepsilon,$$
$$L - \varepsilon < y_n < L + \varepsilon.$$

Note that the upperbound the maximum of two numbers, one needs to upperbound *both numbers involved*. Hence the RHS's of the inequalities above give us

$$\forall n \in \mathbb{Z}_{\geq N} : \max\{x_n, y_n\} < L + \varepsilon.$$

To lowerbound the maximum of two numbers, one needs to lowerbound *at least one of the numbers involved*. The LHS's of the inequalities above in fact are *more than enough* to conclude

$$\forall n \in \mathbb{Z}_{\geq N} : L - \varepsilon < \max\{x_n, y_n\}.$$

Combining these two last inequalities, we have:

$$\forall n \in \mathbb{Z}_{\geq N} : |\max\{x_n, y_n\} - L| < \varepsilon.$$

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$\lrcorner$

7.1 Follow-up Exercises

(i) Here is one way to get more quantitative versions of the above results: Let $x_\bullet : \mathbb{Z}_{\geq 0} \to \mathbb{R}$ be a convergent sequence. The limit of $x_\bullet$ is a uniquely determined real number, so we don't need to explicitly specify it. Then for any $\varepsilon \in \mathbb{R}_{>0}$, the following set of non-negative integers is nonempty:

$$Z_\varepsilon = \left\{ N \in \mathbb{Z}_{\geq 0} \;\middle|\; \forall n \in \mathbb{Z}_{\geq N} : \left| x_n - \lim_{n\to\infty} x_n \right| < \varepsilon \right\}.$$

Then by the Well-Ordering Principle, this set has a unique minimum; let's denote this minimum by $\mathcal{N}_\varepsilon(x_\bullet)$. Thus $\mathcal{N}_\varepsilon(x_\bullet)$ is *exactly* how long one would need to wait for $x_\bullet$ to be within an ε of its limit. We can extend this definition to arbitrary sequences by setting $\mathcal{N}_\varepsilon(x_\bullet) = \infty$ if $Z_\varepsilon = \emptyset$, so that ultimately we have a function

$$\mathcal{N} : \mathbb{R}_{>0} \times F(\mathbb{Z}_{\geq 0}; \mathbb{R}) \to \mathbb{Z}_{\geq 0} \cup \{\infty\}, (\varepsilon, x_\bullet) \mapsto \mathcal{N}_\varepsilon(x_\bullet).$$

Then the above statements can be rephrased in terms of $\mathcal{N}$. For instance, the proof of the statement regarding the limit of the sums of two sequences can be rephrased as:

$$\forall \varepsilon \in \mathbb{R}_{>0} : \mathcal{N}_\varepsilon(x_\bullet + y_\bullet) \leq \max\{\mathcal{N}_{\varepsilon/2}(x_\bullet), \mathcal{N}_{\varepsilon/2}(y_\bullet)\}.$$

How about an *exact* formula for the time on LHS, for this part or the other parts? For instance a bound that seems a priori better is this:

$$\forall \varepsilon \in \mathbb{R}_{>0} : \mathcal{N}_\varepsilon(x_\bullet + y_\bullet) \leq \inf_{\eta \in]0,1[} \max\{\mathcal{N}_{(1-\eta)\varepsilon}(x_\bullet), \mathcal{N}_{\eta\varepsilon}(y_\bullet)\}.$$

(ii) One can generalize the lemma above as follows: Let X be a set, $\alpha, \beta \in$ F(X; $\mathbb{R}$). Then

$$|\sup \alpha - \sup \beta| \leq \sup |\alpha - \beta|.$$

(iii) State and prove versions of the last statement with minima.

8 PSet 3, Problem 3

Let $x_\bullet$ be a sequence of real numbers that converges to zero. Then for any bounded sequence $b_\bullet$ of real numbers,

$$\lim_{n\to\infty} x_n b_n = 0.$$

This claim is indeed true without any modifications:

Claim: $\forall x_\bullet, b_\bullet : \mathbb{Z}_{\geq 0} \to \mathbb{R}$: if $x_\bullet \to 0$ and $\sup_m b_m < \infty$, then $x_\bullet b_\bullet \to 0$. ⌟

Proof: Instead of starting with an $\epsilon > 0$, it's often useful to start from the end, in other words with the expression we are trying to bound. For any $n \in \mathbb{Z}_{\geq 0}$, one has

$$|x_n b_n| = |x_n|\,|b_n| \leq |x_n| \underbrace{\sup_{m\in\mathbb{Z}_{\geq 0}} |b_m|}_{<\infty} \qquad (\star)$$

Put $M = \sup_{m\in\mathbb{Z}_{\geq 0}} |b_m| + 1 \in \mathbb{R}_{\geq 0}$. We know that this is a finite nonnegative number, as the sequence $b_\bullet$ (and consequently the sequence $|b_\bullet|$) is bounded. We added 1 to the supremum of $|b_\bullet|$ to ensure that M is a positive number, so that dividing by M is well-defined[15]. Thus to bound $|x_n b_n|$ from above by ε, we can bound $|x_n|$ by ε/M. Indeed, let $\varepsilon \in \mathbb{R}_{>0}$. Then as $x_\bullet \to 0$, there is an $N \in \mathbb{Z}_{\geq 0}$ such that for any $n \in \mathbb{Z}_{\geq N}$, we have

$$|x_n| = |x_n - 0| < \frac{\varepsilon}{M}.$$

[15] Alternatively, we could have done a case analysis: either $b_\bullet$ is the sequence that is constantly zero, in which case there is nothing to do, or else $b_\bullet$ is not constantly zero, so that the supremum of the absolute value sequence $|b_\bullet|$ is strictly positive.

Here we used the number $\frac{\varepsilon}{M}$ as the "epsilon" in the definition of $x_\bullet$ converging to 0. For any $n \in \mathbb{Z}_{\geq N}$, $(\star)$ also holds; hence combining the two we have that for any $n \in \mathbb{Z}_{\geq N}$:

$$|x_n b_n - 0| = |x_n b_n| < \frac{\varepsilon}{M}(M - 1) < \varepsilon.$$

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8.1 Follow-up Exercises

(i) Let $x_\bullet$ be a sequence of real numbers. Then the following are equivalent:

(a) $x_\bullet$ converges to 0.

(b) For any bounded sequence $b_\bullet$, $x_\bullet b_\bullet$ converges to 0.

(c) For some bounded sequence $b_\bullet$, $x_\bullet b_\bullet$ converges to 0.

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9 PSet 4, Problem 1

{001}

Let I be a (nonempty) interval, $f \in F(I;\mathbb{R})$. Then the following are equivalent:

A: f is continuous.

B: For any $k \in \mathbb{Z}_{\geq 1}$ and any $x_1, \ldots, x_k \in I$, there is an $x_\dagger \in I$ such that $f(x_\dagger) = \frac{f(x_1)+\ldots+f(x_k)}{k}$.

Continuity implies the "averageability" property, but not vice versa in general. For I an interval and $f \in F(I;\mathbb{R})$, let's say that f has the **average value property** if for any finite nonempty subset $S \subseteq \mathrm{im}(f)$, there is an $x_S \in I$ such that

$$f(x_S) = \frac{1}{\#(S)} \sum_{y \in S} y.$$

Similarly let's say that f has the **intermediate value property** if for any subinterval $J \subseteq I$, $f(J)$ is an interval.

Thus the statement **B** above is equivalent to the average value property, and the Intermediate Value Theorem can be concisely stated like so: "The image of any interval under a real valued continuous function is an interval."

(Note that neither of these properties require any assumption on the domain X.)

Claim: Let I be a (nonempty) interval, $f \in F(I;\mathbb{R})$. Then:

(i) If f has the intermediate value property, then it has the average value property.

(ii) If f is continuous, then it has the average value property.

⌟

Proof: The second item follows immediately from the first by way of the Intermediate Value Theorem. Thus it suffices to prove the first item. Suppose f has the intermediate value property, let $S \subseteq \mathrm{im}(f)$ be a finite and nonempty set. Then for any $y \in S$,

$$\min(S) \leq y \leq \max(S).$$

Note that as S is a finite set, it's maximum and minimum are well defined, and consequently are in the image of f. Adding all elements in S, and rearranging, we have

$$\frac{1}{\#(S)} \sum_{y \in S} y \in [\min(S), \max(S)] \subseteq \mathrm{im}(f),$$

hence by the hypothesis that f satisfies the intermediate value property, there is at least one $x_S \in I$ such that

$$f(x_S) = \frac{1}{\#(S)} \sum_{y \in S} y,$$

as was to be shown.

⌟

Here is another proof of the same claim using induction.

Proof (Using Induction)**:** We'll use induction on $\#(S)$. If $\#(S) = 1$, then by definition there is nothing to do. Suppose k_* is a positive integer such that for any subset $S \subseteq \mathrm{im}(f)$ with $\#(S) = k_*$, the statement is true. Let $T \subseteq \mathrm{im}(f)$ be a subset with $\#(T) = k_* + 1$. Let $y_* \in T$ and consider the set $S = T \setminus \{y_*\}$. Then by induction hypothesis, there is an $x_\star \in I$ such that

$$f(x_\star) = \frac{1}{k_*} \sum_{y \in S} y.$$

Then

$$\begin{aligned}
\frac{1}{k_* + 1} \sum_{y \in T} y &= \frac{1}{k_* + 1} \left(\sum_{y \in S} y + y_* \right) \\
&= \frac{1}{k_* + 1} \left(\frac{k_*}{k_*} \sum_{y \in S} y + y_* \right) \\
&= \frac{1}{k_* + 1} \left(k_* f(x_\star) + y_* \right) \\
&= \underbrace{\frac{k_*}{k_* + 1}}_{=1-t} f(x_\star) + \underbrace{\frac{1}{k_* + 1}}_{=t} y_* \\
&= (1 - t) f(x_\star) + t y_*.
\end{aligned}$$

Thus the average over T between two values f attains, hence by the intermediate value property, there is an $x_\dagger \in I$ such that

$$f(x_\dagger) = \frac{1}{k_* + 1} \sum_{y \in T} y.$$

Hence by the principle of mathematical induction, we are done.

┘

For a counterexample then, it suffices to come up with an example of a function that has the intermediate value property *without* being continuous.

Claim: Let I be an interval of positive length. Then there is a function $f \in F(I; \mathbb{R})$ such that

- f is discontinuous, and
- f has the average value property.

⌟

Proof: Let $\lambda, \rho \in \mathrm{I}$, $\lambda \lneq \rho$, and consider

$$\mathrm{f} : \mathrm{I} \to \mathbb{R}, \mathrm{x} \mapsto \begin{cases} \dfrac{\mathrm{x} - \lambda}{\rho - \lambda}, & \text{if } \lambda \leq \mathrm{x} \lneq \rho \\ 0, & \text{else} \end{cases}$$

The image of this function is $[0, 1]$, and hence by the above claim it satisfies the average value property. However this function is discontinuous at ρ.

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9.1 Follow-up Exercises

(i) A subset $A \subseteq \mathbb{R}$ is said to be **closed under averages** if for any finite nonempty $S \subseteq A$, $\frac{1}{\#(S)} \sum_{s \in S} s \in A$. Then

(a) The intersection of $\mathbb{Q}$ with any interval is closed under averages.

(b) Let $A \subseteq [0,1] \cap \mathbb{Q}$. Then A is closed under averages iff $A = [0,1] \cap \mathbb{Q}$.

(ii) A real valued function defined on an interval has the intermediate value property iff it has the average value property.

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10 PSet 5, Problem 1

Let $I = [0, 1]$.

(i) If $f_* \in \mathrm{Homeo}(I)$ and $\varepsilon \in \mathbb{R}_{>0}$, then there is a piecewise linear $f \in \mathrm{Homeo}(I)$ such that $d_{C^0}(f, f_*) < \varepsilon$.

(ii) If $f_\bullet : \mathbb{Z}_{\geq 0} \to \mathrm{Homeo}(I)$ converges uniformly to a function $f_\infty \in F(I; I)$, then $f_\infty \in \mathrm{Homeo}(I)$.

(iii) If $f_* \in \mathrm{Homeo}(I)$, then there is an $\varepsilon \in \mathbb{R}_{>0}$ such that for any $f \in C^0(I; I)$, if $d_{C^0}(f, f_*) < \varepsilon$, then $f \in \mathrm{Homeo}(I)$.

(iv) If $f, g \in \mathrm{Homeo}(I)$ are both two strictly increasing, then there is a $\Phi \in \mathrm{Homeo}(I)$ such that $\Phi \circ f = g \circ \Phi$.

We study the given statements one by one. First we establish a claim regarding the monotonicity property of homeomorphisms of the unit interval, namely that depending on how a given homeomorphism maps the endpoints, we can determine whether or not it is strictly increasing or strictly decreasing.

Claim: Let $f \in \mathrm{Homeo}(I)$. Then

(i) $f(\{0, 1\}) = \{0, 1\}$.

(ii) $f(0) = 0$ iff $f(1) = 1$ iff f is strictly increasing.

(iii) $f(0) = 1$ iff $f(1) = 0$ iff f is strictly decreasing.

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Proof: **(i)** For the first part, suppose wlog $f(0) \in]0, 1[$. Then since f is a bijection and hence an injection, either for any $x \in]0, 1]$, $f(x) > f(0)$ or for any $x \in]0, 1]$, $f(x) < f(0)$. In either case, $\mathrm{im}(f)$ is a proper subset of $[0, 1]$, that is, f is not a surjection, a contradiction.

Thus f preserves the set of endpoints of I. Since f is bijective, there are exactly two cases:

- $f(0) = 0$ and $f(1) = 1$, corresponding to the second claim, and
- $f(0) = 1$ and $f(1) = 0$, which corresponds to the third claim.

(ii) Suppose $f(0) = 0$. Let $x_1, x_2 \in I$ be such that $x_1 < x_2$ but $f(x_1) > f(x_2)$. But then $f(x_2) \in]0, f(x_1)[$, and since the restriction $f|_{[0,x_1]}$ is continuous, there is an $x_3 \in]0, x_1[$ such that $f(x_3) = f(x_2)$, so that f fails to be an injection, a contradiction. Thus f is strictly increasing. Conversely, if f is strictly increasing, then as $0 < 1$, $f(0) < f(1)$, and since both of these are elements in $\{0, 1\}$, it must be the case that $f(0) = 0$.

(iii) The third part follows from the second part as we can replace f with $f \circ \iota$ for ι the homeomorphism defined by $x \mapsto 1 - x$[16].

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Next we'll discuss PL approximation of homeomorphisms of the unit interval. This will help us engineer many concrete homeomorphisms that are fairly straightforward to draw and visualize. Here is some heuristics (♡) as to how this can be accomplished: For f a homeomorphism of the unit interval, we consider the diagonal map $f_0 = \mathrm{id}_I : x \mapsto x$ as a first attempt at a PL approximation. Then the function $|f_0 - f|$ itself is a continuous, non-negative function on I, and hence by the Maximum Principle, it attains its maximum at some point $x_1 \in]0, 1[$; note that this maximum is exactly the C^0 distance from f_0 to f. Now replace f_0 with the PL homeomorphism f_1 whose graph consists of the two line segments connecting any consecutive two of the following three points on the graph of f:

[16] Precomposing a homeomorphism by ι geometrically corresponds to reflecting the graph of the homeomorphism along the line $x = 1/2$.

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$$(0, f(0)), \quad (x_1, f(x_1)), \quad (1, f(1)).$$

Again by the Maximum Principle, there is some point $x_2 \in]0, 1[\setminus\{x_1\}$ where $|f_1 - f|$ is maximized. Again we replace f_1 by the PL homeomorphism f_2 whose graph connects any two consecutive three points (say for simplicity $x_2 > x_1$):

$$(0, f(0)), \quad (x_1, f(x_1)), \quad (x_2, f(x_2)), \quad (1, f(1)).$$

Continuing this way, one ought to be able to approximate f uniformly as accurately as needed. While this method would work, formalization of it is rather cumbersome to write down (without additional groundwork), and we'll instead present a proof that is of a similar flavor, but is less algorithmic[17].

We'll describe the approximating PL homeomorphisms using the notion of a (finite interval) partition of a compact interval. We introduced this notion later than the deadline of this problem set, still partitions are not necessary to solve this problem, instead they are (for me) convenient to use, especially because partitions allow one to suppress indices.

Let f be a homeomorphism of I, and $\mathcal{P}$ be a (finite interval) partition of I with each $P \in \mathcal{P}$ of positive length. We define a new function $f_{\mathcal{P}} : I \to I$ by the condition that for each $P \in \mathcal{P}$, the graph of $f_{\mathcal{P}}$ restricted to P is the line segment from $(\inf(P), f(\inf(P))$ to $(\sup(P), f(\sup(P)))$. A straightforward calculation gives that the explicit formula for $f_{\mathcal{P}}$ is as follows: for any $P \in \mathcal{P}$ and for any $x \in P$,

$$f_{\mathcal{P}} : x \mapsto m_P x + b_P,$$

[17] The algorithm we just described vaguely, while theoretically valid, is not quite practical. For instance with successive steps it is not necessarily true that for any fixed x_0, the vertical difference along the $x = x_0$ is strictly increasing in general. Efficient approximation of functions using simpler functions is the main focus of approximation theory.

where

$$m_P = \frac{f(\sup(P)) - f(\inf(P))}{\sup(P) - \inf(P)},$$
$$b_P = \frac{\sup(P)f(\inf(P)) - \inf(P)f(\sup(P))}{\sup(P) - \inf(P)}.$$

Then by construction $f_{\mathcal{P}}$ is a well defined function from I to I; further it is piecewise linear and continuous. Next note that by the previous established claim f is either strictly increasing or strictly decreasing, which implies that $f_{\mathcal{P}}$ is either strictly increasing or strictly decreasing, hence by a theorem proved in class, $f_{\mathcal{P}}$ is a homeomorphism of I.

Claim: For any $f_* \in \mathrm{Homeo}(I)$ and $\varepsilon \in \mathbb{R}_{>0}$, there is a piecewise linear $f \in \mathrm{Homeo}(I)$ such that

$$d_{C^0}(f, f_*) < \varepsilon.$$

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Proof: Let f_* be a homeomorphism of the interval; wlog we may assume that it is strictly increasing. Let further $\varepsilon \in \mathbb{R}_{>0}$. By a theorem stated (but not proved) in class, f_* is uniformly continuous. Hence there is a $\delta \in \mathbb{R}_{>0}$ such that for any $x_1, x_2 \in I$,

$$|x_1 - x_2| < \delta \implies |f(x_1) - f(x_2)| < \varepsilon/2.$$

By the Archimedean Principle, there is a positive integer n such that $\delta > 1/n$. Let $\mathcal{P}$ be the finite interval partition of I such that each $P \in \mathcal{P}$ has length exactly $1/n$, and consider the associated PL homeomorphism $f_{\mathcal{P}}$. We claim that $f_{\mathcal{P}}$ is less than ε away from f relative to the C^0 metric. Indeed

$$\begin{aligned}
d_{C^0}(f_{\mathcal{P}}, f_*) &= \max_{P\in\mathcal{P}} \max_{\inf(P)\leq x\leq\sup(P)} |f_{\mathcal{P}}(x) - f(x)| \\
&= \max_{P\in\mathcal{P}} \max_{\inf(P)\leq x\leq\sup(P)} |m_P x + b_P - f(x)| \\
&\leq \max_{P\in\mathcal{P}} \max_{\inf(P)\leq x\leq\sup(P)} \operatorname{diam}(f(P)) \\
&\leq \max_{P\in\mathcal{P}} \max_{\inf(P)\leq x\leq\sup(P)} \varepsilon/2 \\
&= \varepsilon/2 < \varepsilon.
\end{aligned}$$

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The remaining three claims are not quite accurate the way they are stated[18].

Claim: A function from I to itself that is the uniform limit of a sequence of homeomorphisms of I need not be a homeomorphism in general.

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Proof: We'll construct a sequence of PL homeomorphisms that converges uniformly to a PL continuous function that fails to be a homeomorphism. For any $n \in \mathbb{Z}_{\geq 1}$, let $f_n : I \to \mathbb{R}$ be the PL continuous function whose graph consists of the three line segments connecting any consecutive two of the following four points:

$$(0,0), \quad \left(\frac{1}{4}, \frac{1}{2} - \frac{1}{n}\right), \quad \left(\frac{3}{4}, \frac{1}{2} + \frac{1}{n}\right), \quad (1,1).$$

Then for any $n \in \mathbb{Z}_{\geq 3}$, f_n is strictly increasing and onto I, hence is a homeomorphism of I. Similarly, let $f_\infty : I \to I$ be the PL continuous function whose graph consists of the three line segments connecting any consecutive two of the following four points:

[18] Still, all of them can be made true with appropriate, and fairly reasonable, modifications. We list some such modifications as follow-up exercises.

$$(0,0), \quad \left(\frac{1}{4}, \frac{1}{2}\right), \quad \left(\frac{3}{4}, \frac{1}{2}\right), \quad (1,1).$$

As f_∞ is not injective, it can not be a homeomorphism. A straightforward calculation shows that for any $n \in \mathbb{Z}_{\geq}3$, the C^0 distance between f_n and f_∞ is exactly $1/n$, hence $f_\bullet \to f_\infty$ uniformly.

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Claim: For any $\varepsilon \in \mathbb{R}_{>0}$, there is an $f_\varepsilon \in C^0(I; I)$ such that

- f_ε is not a homeomorphism, and
- $d_{C^0}(f_\varepsilon, \mathrm{id}_I) < \varepsilon$.

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Proof: For any $n \in \mathbb{Z}_{\geq 3}$, let $f_n : I \to \mathbb{R}$ be the PL continuous function whose graph consists of the three line segments connecting any consecutive two of the following four points:

$$(0,0), \quad \left(\frac{1}{2} - \frac{1}{n}, \frac{1}{2}\right), \quad \left(\frac{1}{2} + \frac{1}{n}, \frac{1}{2}\right), \quad (1,1).$$

None of these functions is injective, hence no f_n is a homeomorphism. By a straightforward calculation, we have that the C^0 distance between f_n and id_I is exactly $1/n$, hence given $\varepsilon \in \mathbb{R}_{>0}$, by the Archimedean Principle we are done.

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Claim: Two strictly increasing homeomorphisms of I need not be topologically conjugate in general.

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Proof: Put $I = [0,1]$ and let $g : [0,1] \to [0,1]$, $x \mapsto x^2$. As for $x_1, x_2 \in I$,

$$|g(x_1) - g(x_2)| = |x_1^2 - x_2^2| = |x_1 + x_2||x_1 - x_2| \leq 2|x_1 - x_2|,$$

g is (Lipschitz hence) continuous. g is further strictly increasing with image I, consequently it is a strictly increasing homeomorphism of the unit interval. $f = \mathrm{id}_I$ too is a strictly increasing homeomorphism of I. If there were a topological conjugacy Φ from f to g, we would have

$$\Phi = \Phi \circ f = g \circ \Phi \implies g = \mathrm{id}_I;$$

this is a contradiction since $g(1/2) = 1/4 \neq 1/2$.

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10.1 Follow-up Exercises

(i) Verify all "straightforward calculations" in the solution presented above.

(ii) Write explicitly a formula for a PL continuous function that interpolates finitely many points.

(iii) Let $f \in \mathrm{Homeo}(I)$ and $\iota \in \mathrm{Homeo}(I)$ be defined by $x \mapsto 1 - x$. Then

(a) f is strictly increasing iff there is a continuous $\gamma : I \to \mathrm{Homeo}(I)$ such that $\gamma(0) = \mathrm{id}_I$ and $\gamma(1) = f$.

(b) f is strictly decreasing iff there is a continuous $\gamma : I \to \mathrm{Homeo}(I)$ such that $\gamma(0) = \iota$ and $\gamma(1) = f$.

(c) f is strictly increasing iff $f \circ \iota$ is strictly decreasing.

(iv) Formalize the heuristics (♡).

(v) For sequences of PL continuous functions, uniform convergence is equivalent to convergence of their control points. More precisely; let I be a compact interval, $m \in \mathbb{Z}_{\geq 1}$,

$$x_{1,\bullet}, x_{2,\bullet}, \cdots, x_{m,\bullet} : \mathbb{Z}_{\geq 0} \cup \{\infty\} \to I,$$
$$y_{1,\bullet}, y_{2,\bullet}, \cdots, y_{m,\bullet} : \mathbb{Z}_{\geq 0} \cup \{\infty\} \to \mathbb{R}$$

be such that for any $n \in \mathbb{Z}_{\geq 0} \cup \{\infty\}$,

$$0 \lneq x_{1,n} \lneq x_{2,n} \lneq \cdots \lneq x_{m,n} \lneq 1.$$

Further, for any $n \in \mathbb{Z}_{\geq 0} \cup \{\infty\}$, let $f_n : I \to \mathbb{R}$ be the PL continuous function whose graph consists of the line segments connecting any consecutive two of the following points:

$$(0,0), \quad (x_{1,n}, y_{1,n}), \quad (x_{2,n}, y_{2,n}), \quad \cdots \quad (x_{m,n}, y_{m,n}), \quad (1,1).$$

Then $f_\bullet \to f_\infty$ uniformly iff for any $i \in \{1, 2, ..., m\}$, $x_{i,\bullet} \to x_{i,\infty}$ and $y_{i,\bullet} \to y_{i,\infty}$.

(vi) Let $f_\bullet : \mathbb{Z}_{\geq 0} \to \mathrm{Homeo}(I)$ be a sequence of functions and $f_\infty \in F(I; I)$. Suppose $f_\bullet \to f_\infty$ uniformly. Then

  (a) If the sequence $f_\bullet^{-1}$ of functional inverses is uniformly convergent, then f_∞ is a homeomorphism and f_∞^{-1} is the uniform limit of $f_\bullet^{-1}$.

  (b) If f_∞ is a homeomorphism, then the sequence $f_\bullet^{-1}$ of functional inverses converges uniformly to f_∞^{-1}.

(vii) For any $f \in \mathrm{Homeo}(I)$ and any $\varepsilon \in \mathbb{R}_{>0}$, there is an $f_\varepsilon \in C^0(I; I)$ such that

- f_ε is not a homeomorphism, and
- $d_{C^0}(f_\varepsilon, f) < \varepsilon$.

(viii) $ð : (f, g) \mapsto d_{C^0}(f, g) + d_{C^0}(f^{-1}, g^{-1})$ is a metric on the collection $\mathrm{Bij}(I)$ of all bijections from I to itself, as well as $\mathrm{Homeo}(I)$.

(ix) Composition of functions is continuous on $F_b(I; I)$, that is, if $f_\bullet \to f_\infty$ and $g_\bullet \to g_\infty$ are two uniformly convergent sequences of bounded functions, then $g_\bullet \circ f_\bullet \to g_\infty \circ f_\infty$ uniformly.

(x) Composition of functions is continuous on $\mathrm{Homeo}(I)$ relative to both d_{C^0} and $ð$.

(xi) Inversion of functions is discontinuous relative to the C^0 metric but continuous relative to ð.

(xii) For any $\alpha \in \mathbb{R}_{>0}$, let $e_\alpha : I \to I$, $x \mapsto x^\alpha$. Then

  (a) $e_\bullet : \mathbb{R}_{>0} \to \mathrm{Homeo}(I)$ is a continuous family of strictly increasing homeomorphisms.

  (b) For any $\alpha, \beta \in \mathbb{R}_{>0}$: $\alpha \neq 1 \neq \beta$ iff e_α and e_β are topologically conjugate.

(xiii) Let $f, g \in \mathrm{Homeo}(I)$. Then

  (a) If f and g are topologically conjugate, then $f(0) = g(0)$.

  (b) If $f(0) = g(0)$ and $\mathrm{Fix}(f) \cap]0, 1[= \emptyset = \mathrm{Fix}(g) \cap]0, 1[$, then f and g are topologically conjugate.

(xiv) Let $f, g \in \mathrm{Homeo}(I)$ be such that $f(0) = g(0)$, and $\Phi \in \mathrm{Homeo}(I)$. Then Φ is a topological conjugacy from f to g iff

  (a) $\Phi(\mathrm{Fix}(f)) = \mathrm{Fix}(g)$ and

  (b) For any interval $J \subseteq I \setminus \mathrm{Fix}(f)$ and for any $x \in [\inf(J), \sup(J)]$, $\Phi \circ f(x) = g \circ \Phi(x)$.

11 PSet 6, Problem 5

Let I be a compact interval, $f_\bullet : \mathbb{Z}_{\geq 0} \to R(I;\mathbb{R})$, $f_\infty \in F(I;\mathbb{R})$.
Consider the following statements:

I: $f_\bullet \to f_\infty$ uniformly.

II: $f_\bullet \to f_\infty$ pointwise.

III: f_∞ is Riemann integrable.

IV: f_∞ is Riemann integrable and

$$\lim_{n\to\infty} {}_R\int_I f_n(x)dx = {}_R\int_I f_\infty(x)dx.$$

V: For each $n \in \mathbb{Z}_{\geq 0}$, $|f_n - f_\infty|$ is Riemann integrable and

$$\lim_{n\to\infty} {}_R\int_I |f_n(x) - f_\infty(x)|dx = 0.$$

Then

(i) If **I**, then **IV** and **V**.

(ii) If **II**, then **III**.

(iii) If **II** and **III**, then **IV**.

(iv) If **II** and **IV**, then **V**.

All but the first statement are false the way they are stated, although they can be modified to obtain correct versions. In a sense, the types of counterexamples we'll present here are exactly the obstructions for correct statements. We start with consequences of uniform convergence.

Claim (I $\implies$ III): Let I be a compact interval, $f_\bullet : \mathbb{Z}_{\geq 0} \to R(I;\mathbb{R})$, $f_\infty \in F(I;\mathbb{R})$. If $f_\bullet \to f_\infty$ uniformly, then $f_\infty \in R(I;\mathbb{R})$.

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Proof: Let $f_\bullet : I \to \mathbb{R}$ be a sequence of Riemann integrable functions converging uniformly to the function $f_\infty : I \to \mathbb{R}$. We claim that f_∞ is Riemann integrable. Let us use the aperture characterization of Riemann integrability. As Darboux apertures are defined only for bounded functions; so we first prove that f_∞ is bounded. By uniform convergence, choosing $\varepsilon = 1$, there is an N such that

$$d_{C^0}(f_N, f_\infty) \leq 1.$$

But then for any $x \in I$, by the triangular inequality we have

$$\begin{aligned} |f_\infty(x)| &\leq |f_\infty(x) - f_N(x)| + |f_N(x)| \\ &\leq d_{C^0}(f_N, f_\infty) + \sup(|f_N|) \\ &\leq 1 + \sup(|f_N|). \end{aligned}$$

f_N is Riemann integrable, hence bounded (a fact we proved in class), hence $\sup(|f_\infty|) \leq 1 + \sup(|f_N|) < \infty$, so that f_∞ too is bounded.

Next we prove that f_∞ is Riemann integrable. Let $\mathcal{P} \in \mathrm{Par}(I)$, $P \in \mathcal{P}$ and $x_1, x_2 \in P$. Then for n arbitrary we have

$$\begin{aligned} &|f_\infty(x_1) - f_\infty(x_2)| \\ &\leq |f_\infty(x_1) - f_n(x_1)| + |f_n(x_1) - f_n(x_2)| + |f_n(x_2) - f_\infty(x_2)| \\ &\leq 2d_{C^0}(f_n, f_\infty) + |f_n(x_1) - f_n(x_2)| \\ &\leq 2d_{C^0}(f_n, f_\infty) + \mathrm{diam}(f_n(P)). \end{aligned}$$

Thus we have

$$\operatorname{diam}(f_\infty(P)) \leq 2d_{C^0}(f_n, f_\infty) + \operatorname{diam}(f_n(P)).$$

(Note that this bound also implies that f_∞ must be bounded, as there are only finitely many elements P in $\mathcal{P}$.)

Taking the Darboux apertures we obtain, for any $\mathcal{P}$ and any n:

$$\begin{aligned} D(f_\infty; \mathcal{P}) &\leq \sum_{P \in \mathcal{P}} [2d_{C^0}(f_n, f_\infty) + \operatorname{diam}(f_n(P))]\ell(P) \\ &= 2d_{C^0}(f_n, f_\infty)\ell(I) + D(f_n; \mathcal{P}). \end{aligned}$$

Let $\varepsilon \in \mathbb{R}_{>0}$. Then by the uniform convergence hypothesis there is an $N = N(\varepsilon)$ such that

$$2d_{C^0}(f_N, f_\infty)\ell(I) < \varepsilon/2.$$

f_N is Riemann integrable, thus by the aperture characterization of Riemann integrability, there is a partition $\mathcal{P} = \mathcal{P}(\varepsilon, N) \in \operatorname{Par}(I)$ such that

$$D(f_N; \mathcal{P}) < \varepsilon/2.$$

Thus for this particular partition $\mathcal{P}$, we have

$$D(f_\infty; \mathcal{P}) \leq 2d_{C^0}(f_N, f_\infty)\ell(I) + D(f_N; \mathcal{P}) < \varepsilon/2 + \varepsilon/2 = \varepsilon.$$

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Now that we have guaranteed the Riemann integrability of the uniform limit of a sequence of Riemann integrable functions, we move on to the convergence of the Riemann integrals.

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Claim (I $\implies$ IV and V): If $f_\bullet \to f_\infty$ uniformly on I and $f_\bullet$ is a sequence of Riemann integrable functions, then

(i) $\int_I |f_\bullet(x) - f_\infty(x)| dx \to 0$.

(ii) $\int_I f_\bullet(x)dx \to \int_I f_\infty(x)dx$.

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Proof: We first note that by an exercise stated in class, for any n, $|f_n - f_\infty|$, is Riemann integrable and further, it suffices to verify the first statement, as for any n,

$$\left|\int_I f_n(x)dx - \int_I f_\infty(x)dx\right| \le \int_I \left|f_n(x) - f_\infty(x)\right| dx.$$

Note that for any $x \in I$, $\left|f_n(x) - f_\infty(x)\right| \le d_{C^0}(f_n, f_\infty)$, hence by the monotonicity of Riemann integrals, we have

$$0 \le \int_I \left|f_n(x) - f_\infty(x)\right| dx \le d_{C^0}(f_n, f_\infty)\ell(I).$$

Taking limits as $n \to \infty$, we are done.

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Next we study the remaining statements[19]. The main idea is that with pointwise convergence, the chain of implications that works in the case of uniform convergence may fail at every single step (without additional assumptions).

Claim (II $\not\Longrightarrow$ III): Let I be a compact interval of positive length. Then there are $f_\bullet : \mathbb{Z}_{\ge 0} \to R(I; \mathbb{R})$, $f_\infty \in F(I; \mathbb{R})$ such that

- $f_\bullet \to f_\infty$ pointwise, but
- $f_\infty \notin R(I; \mathbb{R})$.

[19]As you are reading through the counterexamples, drawing sketches of the functions is recommended.

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Proof: Wlog we may assume $I = [0, 1]$. Define $f_\infty : I \to \mathbb{R}$ by $x \mapsto 1$ if $x \in \mathbb{Q}$ and 0 otherwise. We showed in class that this function is not Riemann integrable. Thus it suffices to construct a sequence of Riemann integrable functions that converges pointwise to f_∞.

To build such a sequence, we first construct a sequence that visits any rational number in I exactly once. Note that

$$I \cap \mathbb{Q} = \bigcup_{b \in \mathbb{Z}_{\geq 1}} \left\{ 0 = \frac{0}{b}, \frac{1}{b}, \frac{2}{b}, \cdots \frac{b-1}{b}, \frac{b}{b} = 1 \right\}.$$

Thus the sequence $\widetilde{q}_\bullet : \mathbb{Z}_{\geq 1} \to I$ defined by

$$\begin{aligned}
&\widetilde{q}_1 = 0/1 = 0, \widetilde{q}_2 = 1/1 = 1,\\
&\widetilde{q}_3 = 0/2 = 0, \widetilde{q}_4 = 1/2, \widetilde{q}_5 = 2/2 = 1,\\
&\widetilde{q}_6 = 0/3 = 0, \widetilde{q}_7 = 1/3, \widetilde{q}_8 = 2/3, \widetilde{q}_9 = 3/3 = 1,\\
&\widetilde{q}_{10} = 0/4 = 0, \widetilde{q}_{11} = 1/4, \widetilde{q}_{12} = 2/4 = 1/2, \widetilde{q}_{13} = 3/4, \widetilde{q}_{14} = 4/4 = 1,\\
&\dots
\end{aligned}$$

has a subsequence $q_\bullet : \mathbb{Z}_{\geq 1} \to I$ that is a bijection onto $I \cap \mathbb{Q}$. For any $n \in \mathbb{Z}_{\geq 1}$, define $f_n : I \to \mathbb{R}$ by

$$x \mapsto \begin{cases} 1, & \text{if } x \in \{q_k \,|\, k \leq n\} \\ 0, & \text{else} \end{cases}.$$

A straightforward calculation gives that for any n, f_n is Riemann integrable with zero integral. Further, for x irrational, $f_n(x) = 0 = f_\infty(x)$ and for x rational, there is a unique $N \in \mathbb{Z}_{\geq 1}$ such that $q_N = x$, hence for any $n \geq N$, $f_n(x) = 1 = f_\infty(x)$, so that $f_\bullet \to f_\infty$ pointwise.

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Claim (**II** and **III** $\not\Longrightarrow$ **IV**)**:** Let I be a compact interval of positive length. Then there are $f_\bullet : \mathbb{Z}_{\geq 0} \to R(I;\mathbb{R})$, $f_\infty \in F(I;\mathbb{R})$ such that

- $f_\bullet \to f_\infty$ pointwise,
- $f_\infty \in R(I;\mathbb{R})$, but
- $\lim_{n\to\infty} {}_R\!\int_I f_n(x)dx \neq {}_R\!\int_I f_\infty(x)dx$.

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Proof: Again we may assume $I = [0, 1]$. Define $f_\infty : I \to \mathbb{R}$, $x \mapsto 0$. Then f_∞ is Riemann integrable with vanishing Riemann integral. We'll construct a sequence of step functions (which are consequently Riemann integrable) that converges pointwise to f_∞, but the numerical sequence their integrals does not converge to 0. Indeed, for $n \in \mathbb{Z}_{\geq 0}$, define $f_n : I \to \mathbb{R}$ to be the step function that is constantly 2^n on the subinterval $\left[1 - \dfrac{1}{2^n}, 1 - \dfrac{1}{2^{n+1}}\right]$ and zero otherwise. Then by a straightforward calculation for each n, the integral of f_n is 1, and consequently the sequence of integrals is constantly 1, and so does not converge to the integral of f_∞.

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Claim (**II** and **IV** $\not\Longrightarrow$ **V**)**:** Let I be a compact interval of positive length. Then there are $f_\bullet : \mathbb{Z}_{\geq 0} \to R(I;\mathbb{R})$, $f_\infty \in F(I;\mathbb{R})$ such that

- $f_\bullet \to f_\infty$ pointwise,
- $f_\infty \in R(I;\mathbb{R})$,
- $\lim_{n\to\infty} {}_R\!\int_I f_n(x)dx = {}_R\!\int_I f_\infty(x)dx$, but
- $\lim_{n\to\infty} {}_R\!\int_I |f_n(x) - f_\infty(x)|dx \neq 0$.

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Proof: We'll modify the previous counterexample to ensure convergence of integrals but not the integrals of the absolute values. Wlog assume $I = [0,1]$. Define $f_\infty : I \to \mathbb{R}$ by $x \mapsto 0$. For any $n \in \mathbb{Z}_{\geq 0}$, let $f_n : I \to \mathbb{R}$ be the step function defined so that it is constantly 2^n on the subinterval $\left[1 - \frac{1}{2^n}, 1 - \frac{3/2}{2^{n+1}}\right]$, constantly -2^n on the subinterval $\left]1 - \frac{3/2}{2^{n+1}}, 1 - \frac{1}{2^{n+1}}\right]$, and zero otherwise. Again by a straightforward calculation, for each n, the integral of f_n is zero, hence the sequence of integrals converges to the integral of the limit function f_∞, however for any n, $|f_n - f_\infty|$ is the n-th step function in the previous example, hence the sequence of integrals of *this* sequence of functions does not converge to zero.

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11.1 Follow-up Exercises

(i) Verify all calculations in the above solutions.

(ii) Write a closed formula for the n-th term of $\widetilde{q}_\bullet$ as well as of $q_\bullet$.

(iii) (Monotone Convergence Theorem for Riemann Integration) Let I be a compact interval, $f_\bullet : \mathbb{Z}_{\geq 0} \to R(I;\mathbb{R})$, $f_\infty \in F(I;\mathbb{R})$. If

- $f_\bullet \to f_\infty$ pointwise,
- $f_\infty \in R(I;\mathbb{R})$, and
- for any $x \in I$, $f_\bullet(x)$ is nondecreasing,

then

- $\lim_{n\to\infty} {}_R\int_I |f_n(x) - f_\infty(x)| dx = 0$, and
- $\lim_{n\to\infty} {}_R\int_I f_n(x) dx = {}_R\int_I f_\infty(x) dx$.

(iv) (Monotone Convergence Theorem for Henstock-Kurzweil Integration) Let I be a compact interval, $f_\bullet : \mathbb{Z}_{\geq 0} \to HK(I;\mathbb{R})$, $f_\infty \in F(I;\mathbb{R})$. If

- $f_\bullet \to f_\infty$ pointwise,
- for any $x \in I$, $n \mapsto f_n(x)$ is nondecreasing,
- $n \mapsto {}_{HK}\int_I f_n(x) dx$ converges to a finite real number,

then

- $f_\infty \in HK(I;\mathbb{R})$, and
- $\lim_{n\to\infty} {}_{HK}\int_I |f_n(x) - f_\infty(x)| dx = 0$, and
- $\lim_{n\to\infty} {}_{HK}\int_I f_n(x) dx = {}_{HK}\int_I f_\infty(x) dx$.

(v) (Bounded Convergence Theorem for Riemann Integration) Let I be a compact interval, $f_\bullet : \mathbb{Z}_{\geq 0} \to R(I;\mathbb{R})$, $f_\infty \in F(I;\mathbb{R})$. If

- $f_\bullet \to f_\infty$ pointwise,
- $f_\infty \in R(I;\mathbb{R})$, and
- $f_\bullet$ is a bounded sequence relative to the C^0 metric,

then

- $\lim_{n\to\infty} {}_R\int_I |f_n(x) - f_\infty(x)| dx = 0$, and
- $\lim_{n\to\infty} {}_R\int_I f_n(x) dx = {}_R\int_I f_\infty(x) dx$.

(vi) (Bounded Convergence Theorem for Henstock-Kurzweil Integration) Let I be a compact interval, $f_\bullet : \mathbb{Z}_{\geq 0} \to HK(I;\mathbb{R})$, $f_\infty \in F(I;\mathbb{R})$. If

- $f_\bullet \to f_\infty$ pointwise, and
- $\inf_n f_n, \sup_n f_n \in HK(I;\mathbb{R})$,

then

- $n \mapsto {}_{HK}\int_I f_n(x) dx$ converges to a finite real number,
- $f_\infty \in HK(I;\mathbb{R})$,
- $\lim_{n\to\infty} {}_{HK}\int_I |f_n(x) - f_\infty(x)| dx = 0$, and
- $\lim_{n\to\infty} {}_{HK}\int_I f_n(x) dx = {}_{HK}\int_I f_\infty(x) dx$.

12 PSet 7, Problem 3

Let $I = [\lambda, \rho]$ for some $\lambda, \rho \in \mathbb{R}$ with $\lambda < \rho$.

(i) For any $f \in R(I; \mathbb{R})$, there is an $x_* \in]\lambda, \rho[$ such that

$$f(x_*) = \frac{1}{\ell(I)} \int_I f(x)dx.$$

(ii) For any $f \in C^0(I; \mathbb{R})$, there is an $x_* \in]\lambda, \rho[$ such that

$$f(x_*) = \frac{1}{\ell(I)} \int_I f(x)dx.$$

(iii) For any $f \in C^0(I; \mathbb{R})$ and for any $g \in R(I; \mathbb{R})$, there is an $x_* \in]\lambda, \rho[$ such that

$$\int_I f(x)g(x)dx = f(x_*) \int_I g(x)dx.$$

(iv) For any $f \in C^0(I; \mathbb{R})$ and for any $g \in R(I; \mathbb{R}_{\geq 0})$, there is an $x_* \in]\lambda, \rho[$ such that

$$\int_I f(x)g(x)dx = f(x_*) \int_I g(x)dx.$$

This problem on the one hand connects to PSet 4, Problem 1 above and on the other hand to the mean value theorem(s) in differential calculus. Indeed, the kinds of statements given in this problem are often called mean value theorems for integral calculus. The first and third statements are not quite correct, and indeed we may consider the second and fourth statements as corrected versions of them, respectively. Note further that the constant function $x \mapsto 1$ is Riemann integrable, and if in the fourth statement we take g to be this function, we recover the second statement. Thus after verifying that the first and third statements are incorrect the way they are stated, it will suffice to verify the fourth statement.

Claim: Let $I = [\lambda, \rho]$ for some $\lambda, \rho \in \mathbb{R}$ with $\lambda < \rho$. Then there is a function $f \in F(I; \mathbb{R})$ such that

(i) f is a step function (hence is Bourbaki, hence Riemann integrable),

(ii) The integral (hence average) of f is 0,

(iii) 0 is not in the image of f.

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Proof: Let $\mu = (\lambda + \rho)/2$ be the midpoint of I, and define $f : I \to \mathbb{R}$ by $x \mapsto -1$ if $x \leq \mu$ and $x \mapsto 1$ otherwise. Then f is a step function, hence it is Bourbaki and Riemann integrable; its integral and average is zero, but it never attains the value zero.

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Next we study the third statement. It is not quite correct the way it is given, and we'll break it by constructing two functions f, g such that fg has nonzero integral but g has zero integral. For g we'll take the step function we used before.

Claim: Let $I = [\lambda, \rho]$ for some $\lambda, \rho \in \mathbb{R}$ with $\lambda < \rho$. Then there are functions $f, g \in F(I; \mathbb{R})$ such that

(i) f is continuous,

(ii) g is a step function (hence Riemann integrable),

(iii) The integral of g is zero,

(iv) The integral of fg is not zero[20].

⌟

Proof: Let $\mu = (\lambda + \rho)/2$ be the midpoint of I, and define g : I $\to \mathbb{R}$ by x $\mapsto -1$ if x $\leq \mu$ and x $\mapsto$ 1 otherwise. Then g is a step function, hence it is Bourbaki and Riemann integrable. Further the integral of g is zero.

Next, define f : I $\to \mathbb{R}$ by x $\mapsto$ 0 if x $\leq \mu$ and x $\mapsto$ x otherwise. Then f is a continuous function, and

$$\int_I f(x)g(x)dx = \int_{[\mu,\rho]} xdx \gneq 0.$$

⌟

Note that in the proof above, in the last step all we needed was to ensure that the integral of the product is nonzero. For the specific example we had, we could have approximated the integral from below via the integral of a step function, or we could have referred to the following lemma. This lemma is not really necessary, although it is of separate interest. Throughout we'll use various properties of Riemann integrals given as an exercise in class.

Lemma: Let ϕ : I $\to \mathbb{R}$ be a Riemann integrable function defined on a closed and bounded interval I = $[\lambda, \rho]$ with positive length. If

- for any x $\in$ I, ϕ(x) $\gneq$ 0 and
- there is some point $\mu \in$ I where ϕ is continuous[21],

[20]Note that any continuous function is Bourbaki integrable by a problem in a previous problem set, Bourbaki integrability implies Riemann integrability by an exercise given in class, and the product of two Riemann integrable functions is Riemann integrable by a discussion in class.

[21]This assumption actually is redundant; indeed, the Lebesgue-Vitali Theorem states that a bounded function defined on a closed and bounded interval is Riemann integrable iff it is continuous off a "negligible" set. Consequently any Riemann integrable function must be continuous at least at some point. The reason we state the lemma with this additional redundant assumption is so that we don't refer to this more advanced result (which for the purposes of this problem is not needed).

then $\int_I \phi(x)dx \gneq 0$.

⌟

Proof: Since ϕ is continuous at μ, for $\varepsilon \in]0, g(\mu)[$, there is a closed subinterval $J \subseteq I$ containing μ of positive length such that and for any $x \in J$, $|\phi(x) - \phi(\mu)| < \varepsilon$. Thus

$$\forall x \in J : 0 \lneq \phi(\mu) - \varepsilon \lneq \phi(x).$$

The restriction of ϕ to the subinterval J too is Riemann integrable, and by the monotonicity and interval additivity of Riemann integrals we have

$$\begin{aligned}
0 &\lneq \ell(J)(\phi(\mu) - \varepsilon) \\
&= \int_J (\phi(\mu) - \varepsilon)dx \\
&\leq \int_J \phi(x)dx \\
&= \int_{[\lambda,\min(J)]} 0dx + \int_J \phi(x)dx + \int_{[\max(J),\rho]} 0dx \\
&\leq \int_{[\lambda,\min(J)]} \phi(x)dx + \int_J \phi(x)dx + \int_{[\max(J),\rho]} \phi(x)dx \\
&= \int_I \phi(x)dx.
\end{aligned}$$

⌟

Finally we study the remaining two statements:

Claim (MVT for Integral Calculus)**:** Let $I = [\lambda, \rho]$ for some $\lambda, \rho \in \mathbb{R}$

For completeness; here is what it means for a set to be negligible in this context: Let $S \subseteq I$ be a subset and consider the characteristic function of S defined by $\chi_S : I \to \mathbb{R}$, $x \mapsto 1$ if $x \in S$ and $x \mapsto 0$ otherwise. Then S is called **negligible** (or **of zero Lebesgue measure**) if χ_S is HK integrable and its HK integral is zero. Just to give an example, we proved in class that $\mathbb{Q} \cap [0, 1]$ is a negligible subset of $[0, 1]$.

with $\lambda < \rho$. Then for any $f \in C^0(I;\mathbb{R})$:

(i) For any $g \in R(I;\mathbb{R}_{\geq 0})$, there is an $x_* \in]\lambda, \rho[$ such that

$$f(x_*) \int_I g(x)dx = \int_I f(x)g(x)dx. \qquad (\blacklozenge)$$

(ii) There is an $x_* \in]\lambda, \rho[$ such that

$$f(x_*) = \frac{1}{\ell(I)} \int_I f(x)dx.$$

⌟

Proof: As mentioned above, it suffices to show the first item, as choosing $g : x \mapsto 1$ in the first item gives the second item.

For any $x \in I$, we have

$$\min(f) \leq f(x) \leq \max(f).$$

As $g \geq 0$, we may multiply these inequalities by $g(x)$ to obtain

$$\min(f)g(x) \leq f(x)g(x) \leq \max(f)g(x).$$

Integrating, we obtain

$$\begin{aligned} \min(f) \int_I g(x)dx &= \int_I \min(f)g(x)dx \\ &\leq \int_I f(x)g(x)dx \\ &\leq \int_I \max(f)g(x)dx = \max(f) \int_I g(x)dx \end{aligned}$$

Either $\int_I g(x)dx = 0$, xor $\int_I g(x)dx \gneq 0$. In the first case, both sides of $(\blacklozenge)$ are zero, and hence we may take as x_* any point in $]\lambda, \rho[$.

Suppose $\int_I g(x)dx \gneq 0$; then dividing the terms by this positive number, we obtain

$$\min(f) \leq \frac{\int_I f(x)g(x)dx}{\int_I g(x)dx} \leq \max(f).$$

As f is continuous, by the Intermediate Value Theorem, there is an $x_\dagger \in [\lambda, \rho]$ such that

$$f(x_\dagger) = \frac{\int_I f(x)g(x)dx}{\int_I g(x)dx}.$$

If $x_\dagger \in]\lambda, \rho[$, we are done as we can choose $x_* = x_\dagger$. Otherwise, suppose $x_\dagger \in \{\lambda, \rho\}$. If f is a constant function, then again we are done (as any point can be taken as x_*). Thus suppose f is not constant. We claim that for some $y_\dagger, z_\dagger \in]\lambda, \rho[$,

$$f(y_\dagger) \leq \frac{\int_I f(x)g(x)dx}{\int_I g(x)dx} \leq f(z_\dagger).$$

Suppose otherwise, then for $M = \frac{\int_I f(x)g(x)dx}{\int_I g(x)dx}$, $y \mapsto f(y) - M$ is a continuous function defined on $[\lambda, \rho]$ that is *strictly* positive on $]\lambda, \rho[$. By the Maximum Principle, then, either λ or ρ must be a minimum point for this function, where in fact the value attained is zero. Let $\varepsilon \in \mathbb{R}_{>0}$ be less than $\ell(I)/2$. Integrating against g, we obtain

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$$
\begin{aligned}
0 &= \int_I f(y)g(y)dy - \int_I f(x)g(x)dx \\
&= \int_I f(y)g(y)dy - \frac{\int_I f(x)g(x)dx}{\int_I g(x)dx} \int_I g(y)dy \\
&= \int_I f(y)g(y)dy - M \int_I g(y)dy \\
&= \int_I (f(y) - M)g(y)dy \\
&= \underbrace{\int_{[\lambda,\lambda+\varepsilon]} (f(y) - M)g(y)dy}_{\geq 0} \\
&\qquad + \int_{[\lambda+\varepsilon,\rho-\varepsilon]} (f(y) - M)g(y)dy \\
&\qquad\qquad + \underbrace{\int_{[\rho-\varepsilon,\rho]} (f(y) - M)g(y)dy}_{\geq 0} \\
&\geq \int_{[\lambda+\varepsilon,\rho-\varepsilon]} (f(y) - M)g(y)dy \\
&\geq \underbrace{\left(\min_{y\in[\lambda+\varepsilon,\rho-\varepsilon]} f(y) - M\right)}_{\gneq 0} \underbrace{\int_{[\lambda+\varepsilon,\rho-\varepsilon]} g(y)dy}_{\geq 0}.
\end{aligned}
$$

Thus we have that for any $\varepsilon \in \mathbb{R}_{>0}$ with $\varepsilon < \ell(I)/2$,

$$\int_{[\lambda+\varepsilon,\rho-\varepsilon]} g(y)dy = 0.$$

On the other hand, for any such ε,

https://alpuzman.github.io/
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$$\int_I g(x)dx = \int_I g(x)dx - \underbrace{\int_{[\lambda+\varepsilon,\rho-\varepsilon]} g(y)dy}_{=0}$$
$$= \int_{[\lambda,\lambda+\varepsilon]} g(y)dy + \int_{[\rho-\varepsilon,\rho]} g(y)dy$$
$$\leq 2\varepsilon \sup(g).$$

As g is Riemann integrable, it is bounded and hence its supremum is a finite real number, and hence taking the infimum over ε, we obtain that the integral of g vanishes, a contradiction.

A similar argument gives the existence of a $z_\dagger$ as claimed. If $f(y_\dagger) = \frac{1}{\ell(I)} \int_I f(x)dx$ or $\frac{1}{\ell(I)} \int_I f(x)dx = f(z_\dagger)$, we are done. Otherwise again applying the Intermediate Value Theorem to the restriction of f to the closed and bounded interval with endpoints $y_\dagger$ and $z_\dagger$ guarantees the existence of an x_* as claimed.

$\lrcorner$

12.1 Follow-up Exercises

(i) MVT for Integral Calculus can be proved using the Fundamental Theorem(s) of Calculus and (Cauchy's) MVT.

(ii) Let $I = [\lambda, \rho]$ for some $\lambda, \rho \in \mathbb{R}$ with $\lambda < \rho$. Then for any $f \in C^0(I; \mathbb{R})$ and for any $g \in HK(I; \mathbb{R}_{\geq 0})$,

(a) $fg \in HK(I; \mathbb{R})$, and

(b) there is an $x_* \in]\lambda, \rho[$ such that

$$f(x_*) \int_I g(x)dx = \int_I f(x)g(x)dx.$$